Assignment 3:
Collaborative Filtering

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Assignment: Assignment 3 — Collaborative Filtering

1. Introduction

Collaborative Filtering is a widely used recommendation system technique that predicts a user's preference for an item based on the preferences of similar users or similar items. It relies on the assumption that users who agreed in the past will agree in the future, and that they will like similar items to those they liked in the past.

This assignment applies Collaborative Filtering to a movie rating dataset containing four users and four movies. One rating is missing — Alice's rating for The Matrix — and the goal is to predict this missing value using two approaches:

1. User-Based Collaborative Filtering: predict the missing rating by finding users similar to Alice using Cosine Similarity, then using their Matrix ratings as a weighted prediction.
2. Item-Based Collaborative Filtering: predict the missing rating by finding movies similar to The Matrix using Cosine Similarity, then using Alice's ratings for those similar movies.

The Cosine Similarity formula used throughout this assignment is taken directly from the Data Mining B141 lecture by Prof. Dr. Mahmoudreza Babaei:

```
Cos(A, B) = (A . B) / (|A| x |B|)
          = SUM(Ai x Bi) / (sqrt(SUM(Ai^2)) x sqrt(SUM(Bi^2)))
```

2. Dataset

The dataset contains ratings from four users for four movies on a scale of 1 (lowest) to 5 (highest). Alice's rating for The Matrix is missing and is the target of prediction.

User	Inception	Interstellar	The Matrix	The Avengers
Alice	5	4	? (Missing)	1
Bob	4	5	2	1
Charlie	1	2	5	4
David	5	4	1	2

Table 1: Movie Rating Dataset — Alice's Matrix Rating is Missing

The missing value is represented by a question mark (?). In both parts of this assignment, only the known ratings are used when computing similarities involving Alice.

3. Part 1: User-Based Collaborative Filtering

User-Based Collaborative Filtering predicts a missing rating by identifying users who are most similar to the target user (Alice) and using their ratings for the missing item as a weighted prediction.

3.1 Approach

Since Alice's Matrix rating is missing, we use only the three movies she has rated (Inception, Interstellar, The Avengers) to compute similarity between Alice and each other user. The known rating vectors are:

```
Alice   = [5, 4, 1]  (Inception=5, Interstellar=4, Avengers=1)
Bob     = [4, 5, 1]  (Inception=4, Interstellar=5, Avengers=1)
Charlie = [1, 2, 4]  (Inception=1, Interstellar=2, Avengers=4)
David   = [5, 4, 2]  (Inception=5, Interstellar=4, Avengers=2)
```

3.2 Step 1: Cosine Similarity — Alice vs Bob

```
Cos(Alice, Bob) = (A . B) / (|A| x |B|)
```

Dot product:

```
A . B = (5x4) + (4x5) + (1x1)
      = 20 + 20 + 1 = 41
```

Magnitudes:

```
|Alice| = sqrt(5^2 + 4^2 + 1^2)
        = sqrt(25 + 16 + 1)
        = sqrt(42) = 6.4807
|Bob|   = sqrt(4^2 + 5^2 + 1^2)
        = sqrt(16 + 25 + 1)
        = sqrt(42) = 6.4807
```

Cosine Similarity:

```
Cos(Alice, Bob) = 41 / (6.4807 x 6.4807)
                = 41 / 42.0000
                = 0.9762
```

3.3 Step 2: Cosine Similarity — Alice vs Charlie

```
Cos(Alice, Charlie) = (A . C) / (|A| x |C|)
```

Dot product:

```
A . C = (5x1) + (4x2) + (1x4)
      = 5 + 8 + 4 = 17
```

Magnitudes:

```
|Alice|   = sqrt(42) = 6.4807  (same as above)
|Charlie| = sqrt(1^2 + 2^2 + 4^2)
          = sqrt(1 + 4 + 16)
          = sqrt(21) = 4.5826
```

Cosine Similarity:

```
Cos(Alice, Charlie) = 17 / (6.4807 x 4.5826)
                    = 17 / 29.6985
                    = 0.5724
```

3.4 Step 3: Cosine Similarity — Alice vs David

```
Cos(Alice, David) = (A . D) / (|A| x |D|)
```

Dot product:

```
A . D = (5x5) + (4x4) + (1x2)
      = 25 + 16 + 2 = 43
```

Magnitudes:

```
|Alice| = sqrt(42) = 6.4807  (same as above)
|David| = sqrt(5^2 + 4^2 + 2^2)
        = sqrt(25 + 16 + 4)
        = sqrt(45) = 6.7082
```

Cosine Similarity:

```
Cos(Alice, David) = 43 / (6.4807 x 6.7082)
                  = 43 / 43.4741
                  = 0.9891
```

3.5 Similarity Summary

User Pair	Dot Product	|Alice|	|Other User|	Cosine Similarity
Alice vs Bob	41	6.4807	6.4807	0.9762
Alice vs Charlie	17	6.4807	4.5826	0.5724
Alice vs David	43	6.4807	6.7082	0.9891

Table 2: User-Based Cosine Similarity Results

David is most similar to Alice (0.9891), followed closely by Bob (0.9762). Charlie shows much lower similarity (0.5724) — this makes intuitive sense as Charlie rated Inception and Interstellar very low (1 and 2) while Alice rated them very high (5 and 4), indicating opposite tastes.

3.6 Step 4: Predict Alice's Matrix Rating

The weighted average prediction formula uses each user's similarity to Alice as their weight:

```
Predicted Rating = SUM(sim(Alice,u) x rating(u,Matrix))
                   ────────────────────────────────────
                          SUM(|sim(Alice,u)|)
```

Known Matrix ratings of other users: Bob=2, Charlie=5, David=1

```
Numerator = (0.9762 x 2) + (0.5724 x 5) + (0.9891 x 1)
          = 1.9524 + 2.8621 + 0.9891
          = 5.8036
Denominator = |0.9762| + |0.5724| + |0.9891|
            = 2.5377
Prediction = 5.8036 / 2.5377 = 2.2869
Rounded:     2
```

User-Based Predicted Rating for Alice — The Matrix = 2

4. Part 2: Item-Based Collaborative Filtering

Item-Based Collaborative Filtering predicts a missing rating by identifying movies most similar to the target movie (The Matrix) and using Alice's ratings for those similar movies as a weighted prediction.

4.1 Approach

We represent each movie as a vector of ratings from all users who have rated both that movie and The Matrix. Since Alice's Matrix rating is missing, we use only Bob, Charlie, and David's ratings to build the item vectors. Alice's known ratings for other movies are then used in the prediction step.

Movie	Bob	Charlie	David
Inception	4	1	5
Interstellar	5	2	4
The Avengers	1	4	2
The Matrix	2	5	1

Table 3: Item Vectors (Bob, Charlie, David Ratings Only)

4.2 Step 1: Cosine Similarity — The Matrix vs Inception

```
Matrix    = [2, 5, 1]  (Bob=2, Charlie=5, David=1)
Inception = [4, 1, 5]  (Bob=4, Charlie=1, David=5)
```

Dot product:

```
Matrix . Inception = (2x4) + (5x1) + (1x5)
                   = 8 + 5 + 5 = 18
```

Magnitudes:

```
|Matrix|    = sqrt(2^2 + 5^2 + 1^2)
            = sqrt(4 + 25 + 1)
            = sqrt(30) = 5.4772
|Inception| = sqrt(4^2 + 1^2 + 5^2)
            = sqrt(16 + 1 + 25)
            = sqrt(42) = 6.4807
```

Cosine Similarity:

```
Cos(Matrix, Inception) = 18 / (5.4772 x 6.4807)
                       = 18 / 35.4965
                       = 0.5071
```

4.3 Step 2: Cosine Similarity — The Matrix vs Interstellar

```
Matrix       = [2, 5, 1]
```

```
Interstellar = [5, 2, 4]  (Bob=5, Charlie=2, David=4)
```

Dot product:

```
Matrix . Interstellar = (2x5) + (5x2) + (1x4)
                      = 10 + 10 + 4 = 24
```

Magnitudes:

```
|Matrix|       = sqrt(30) = 5.4772  (same as above)
|Interstellar| = sqrt(5^2 + 2^2 + 4^2)
               = sqrt(25 + 4 + 16)
               = sqrt(45) = 6.7082
```

Cosine Similarity:

```
Cos(Matrix, Interstellar) = 24 / (5.4772 x 6.7082)
                          = 24 / 36.7423
                          = 0.6532
```

4.4 Step 3: Cosine Similarity — The Matrix vs The Avengers

```
Matrix   = [2, 5, 1]
Avengers = [1, 4, 2]  (Bob=1, Charlie=4, David=2)
```

Dot product:

```
Matrix . Avengers = (2x1) + (5x4) + (1x2)
                  = 2 + 20 + 2 = 24
```

Magnitudes:

```
|Matrix|   = sqrt(30) = 5.4772  (same as above)
|Avengers| = sqrt(1^2 + 4^2 + 2^2)
           = sqrt(1 + 16 + 4)
           = sqrt(21) = 4.5826
```

Cosine Similarity:

```
Cos(Matrix, Avengers) = 24 / (5.4772 x 4.5826)
                      = 24 / 25.0998
                      = 0.9562
```

4.5 Similarity Summary

Movie Pair	Dot Product	|Matrix|	|Other Movie|	Cosine Similarity
Matrix vs Inception	18	5.4772	6.4807	0.5071

Movie Pair	Dot Product	|Matrix|	|Other Movie|	Cosine Similarity
Matrix vs Interstellar	24	5.4772	6.7082	0.6532
Matrix vs Avengers	24	5.4772	4.5826	0.9562

Table 4: Item-Based Cosine Similarity Results

The Avengers is most similar to The Matrix (0.9562) — both are action/superhero genre films. Interstellar shows moderate similarity (0.6532) and Inception the lowest (0.5071). This aligns with genre intuition: The Matrix and The Avengers share more rating patterns among users who prefer action films over cerebral science fiction.

4.6 Step 4: Predict Alice's Matrix Rating

The weighted average uses each movie's similarity to The Matrix as the weight, combined with Alice's known ratings for those movies:

```
Predicted Rating = SUM(sim(Matrix,i) x Alice_rating(i))
                   ------------------------------------
                          SUM(|sim(Matrix,i)|)
```

Alice's known ratings: Inception=5, Interstellar=4, Avengers=1

```
Numerator = (0.5071 x 5) + (0.6532 x 4) + (0.9562 x 1)
          = 2.5355 + 2.6128 + 0.9562
          = 6.1044
Denominator = |0.5071| + |0.6532| + |0.9562|
            = 2.1165
Prediction = 6.1044 / 2.1165 = 2.8842
Rounded:     3
```

Item-Based Predicted Rating for Alice — The Matrix = 3

5. Comparison and Discussion

5.1 Results Summary

Method	Predicted Rating (Exact)	Predicted Rating (Rounded)
User-Based Collaborative Filtering	2.2869	2
Item-Based Collaborative Filtering	2.8842	3

Table 5: Final Predicted Ratings Comparison

5.2 Why the Predictions Differ

The two methods produce different predictions (2 vs 3) because they use fundamentally different information to make the recommendation:

User-Based approach relies on similar users' actual Matrix ratings. David (most similar, 0.9891) and Bob (0.9762) both rated The Matrix very low (1 and 2 respectively). Charlie, who is least similar to Alice (0.5724), is the only one who rated The Matrix highly (5). Since Charlie carries less weight, the weighted average pulls toward 2.

Item-Based approach relies on how similar The Matrix is to other movies, combined with Alice's own ratings of those movies. The Matrix is most similar to The Avengers (0.9562), which Alice rated 1. But Alice rated Inception (5) and Interstellar (4) much higher, and both have moderate similarity to The Matrix (0.5071 and 0.6532). The higher weights on Alice's high ratings pull the prediction toward 3.

5.3 Which Prediction is More Reliable?

In this small dataset, the Item-Based prediction (3) is arguably more reliable because:

3. It uses Alice's own rating preferences directly — rather than inferring from other users whose tastes may differ in subtle ways.
4. The high similarity between The Matrix and The Avengers (0.9562) is very strong — this is the most reliable single piece of similarity evidence in the dataset.
5. With only 3 users in the similarity pool (after excluding Alice), user-based similarity is more susceptible to noise from individual preference differences.

5.4 Limitations of This Dataset

This dataset has only 4 users and 4 movies — far too small for reliable collaborative filtering in practice. With so few users, a single outlier (Charlie's opposite preferences) significantly influences the user-based prediction. Real recommendation systems typically require thousands of users and items to produce stable, reliable similarity scores.

6. Conclusion

This assignment applied two Collaborative Filtering techniques to predict Alice's missing rating for The Matrix using Cosine Similarity.

In Part 1 (User-Based), Alice was found to be most similar to David (0.9891) and Bob (0.9762), and least similar to Charlie (0.5724). Using the weighted average of their Matrix ratings, the predicted rating for Alice was 2.2869, rounded to 2.

In Part 2 (Item-Based), The Matrix was found to be most similar to The Avengers (0.9562) and moderately similar to Interstellar (0.6532) and Inception (0.5071). Using Alice's own ratings for these movies as weighted inputs, the predicted rating was 2.8842, rounded to 3.

Both methods apply the same Cosine Similarity formula taught in the Data Mining B141 lecture, but use it from different perspectives — one comparing users, the other comparing items. The difference in final predictions (2 vs 3) illustrates that the two approaches capture complementary aspects of user preference and item characteristic, and in practice both are often combined in hybrid recommendation systems.